

Decent Coaching Classes

IX CBSE

Mind Your Math - VII

Answer Key & Marking Scheme

Marks: 80

Time: 3 Hrs

SECTION – A

20 × 1 = 20 marks

1. (B) II — Negative x, positive y → second quadrant [1]
2. (D) (0, -5) — $x = 0$ and $y = -5$ [1]
3. (B) 4 — $3x - 12 = 0 \rightarrow x = 4$ [1]
4. (C) (0, 1) — Put $x = 0 \rightarrow y = 1$ [1]
5. (D) $\sqrt{7}$ — $\sqrt{16} = 4$, $22/7$ and $0.333...$ are rational; 7 is not a perfect square [1]
6. (A) terminating — $7/8 = 0.875$ [1]
7. (B) 4 — $\sqrt{16} = 4$ [1]
8. (C) 9991 — $(100 + 3)(100 - 3) = 10000 - 9 = 9991$ [1]
9. (B) 25 — $49 - 2 \times 12 = 25$ [1]
10. (C) 7 — $9 - 2 = 7$ [1]
11. (B) at equal distances — Equal chords are equidistant from the centre [1]
12. (A) 50° — Half of the angle at the centre [1]
13. (B) 3 cm — $\sqrt{(25 - 16)} = 3$ [1]
14. (A) 6 cm^2 — $s = 6$; $\sqrt{(6 \times 3 \times 2 \times 1)} = 6$ [1]
15. (B) 7 cm — $2 \times 22/7 \times r = 44 \rightarrow r = 7$ [1]
16. (C) $4\sqrt{3} \text{ cm}^2$ — $(\sqrt{3}/4) \times 16 = 4\sqrt{3}$ [1]
17. (B) 1 — A sure event always occurs [1]
18. (D) $1/3$ — Favourable 5, 6 → $2/6 = 1/3$ [1]
19. (C) 19 — Common difference 4 → $15 + 4 = 19$ [1]
20. (B) 55 — $10 \times 11 / 2 = 55$ [1]

SECTION – B

5 × 2 = 10 marks

21. Both points have $x = 2$, so AB is a vertical segment. [$\frac{1}{2}$]
 $AB = |3 - (-5)| = 8$ units [1]
AB is parallel to the y-axis. [$\frac{1}{2}$]
22. Let $x = 0.3636...$ Then $100x = 36.3636...$ [$\frac{1}{2}$]
 $100x - x = 36$, so $99x = 36$ [1]
 $x = 36/99 = 4/11$ [$\frac{1}{2}$]
23. $8a^3 + 27b^3 = (2a)^3 + (3b)^3$ [$\frac{1}{2}$]
Using $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$ [$\frac{1}{2}$]
 $= (2a + 3b)(4a^2 - 6ab + 9b^2)$ [1]
24. Total outcomes = 8 [$\frac{1}{2}$]
(i) $P(\text{red}) = 3/8$ [$\frac{1}{2}$]
(ii) $P(\text{not red}) = 1 - 3/8 = 5/8$ [1]
25. $a = 7$, $d = 3$, $n = 15$; $t_n = a + (n - 1)d$ [1]
 $t_{15} = 7 + 14 \times 3 = 49$ [1]

SECTION – C**6 × 3 = 18 marks**

26. $p(2) = 0$, so $2k - 6 = 0$ [1]
 $k = 3$ [1]
 $p(x) = 3x - 6$, so $p(5) = 15 - 6 = 9$ [1]
27. Multiply and divide by the conjugate $(\sqrt{5} + \sqrt{2})$ [1]
Denominator: $(\sqrt{5})^2 - (\sqrt{2})^2 = 5 - 2 = 3$ [1]
Result = $(\sqrt{5} + \sqrt{2})/3$ [1]
28. Use $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ with $a = 2x$, $b = -3y$, $c = z$ [1]
 $= 4x^2 + 9y^2 + z^2 + 2(2x)(-3y) + 2(-3y)(z) + 2(z)(2x)$ [1]
 $= 4x^2 + 9y^2 + z^2 - 12xy - 6yz + 4zx$ [1]
29. The perpendicular from the centre bisects the chord: $AM = 8$ cm; $OA = 10$ cm [1]
 $OM^2 = OA^2 - AM^2 = 100 - 64 = 36$ [1]
 $OM = 6$ cm [1]
30. $s = (13 + 14 + 15)/2 = 21$ cm [1]
Area = $\sqrt{[s(s-a)(s-b)(s-c)]} = \sqrt{(21 \times 8 \times 7 \times 6)}$ [1]
 $= \sqrt{7056} = 84$ cm² [1]
31. $a = 3$, $d = 4$, $n = 20$ [1]
 $S_n = n/2 [2a + (n-1)d] = 10 [6 + 76]$ [1]
 $= 10 \times 82 = 820$ [1]

SECTION – D**4 × 5 = 20 marks**

32. (a) $2 \times (-3) = -6$; factors of -6 with sum -5 are -6 and 1 [1]
 $2x^2 - 6x + x - 3 = 2x(x-3) + 1(x-3) = (x-3)(2x+1)$ [1]
(b) $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$ [1]
 $81 = a^2 + b^2 + c^2 + 52$ [1]
 $a^2 + b^2 + c^2 = 29$ [1]
33. Given: chord AB, centre O, $OM \perp AB$. To prove: $AM = MB$. [1]
Proof: In right triangles OMA and OMB, $OA = OB$ (radii), $OM = OM$ (common), $\angle OMA = \angle OMB = 90^\circ$. So $\triangle OMA \cong \triangle OMB$ (RHS). [1]
Hence $AM = MB$ (corresponding parts), i.e. M is the midpoint of AB. [1]
Numerical: $AM = 12$ cm, $OA = 13$ cm; $OM^2 = 169 - 144 = 25$ [1]
 $OM = 5$ cm [1]
34. (i) Perimeter = $3 + 4 + 5 + 6 = 18$ cm [1]
(ii) In right $\triangle ABC$, $AC^2 = 3^2 + 4^2 = 25$, so $AC = 5$ cm [1]
(iii) Area of $\triangle ABC = \frac{1}{2} \times 3 \times 4 = 6$ cm² [1]
 $\triangle ACD$ has sides 5, 5, 6: $s = 8$; area = $\sqrt{(8 \times 3 \times 3 \times 2)} = \sqrt{144} = 12$ cm² [1]
Area of ABCD = $6 + 12 = 18$ cm² [1]
35. (i) $AB = 5 - 1 = 4$ units; $BC = 4 - 1 = 3$ units [2]
(ii) Area = $4 \times 3 = 12$ sq. units [1]
Perimeter = $2(4 + 3) = 14$ units [1]
(iii) The diagonals bisect each other: midpoint of AC = $((1+5)/2, (1+4)/2) = (3, 2.5)$ [1]

SECTION – E**3 × 4 = 12 marks**

36. (i) $P(\text{cricket}) = 16/40 = 2/5$ [1]
(ii) $P(\text{football or kabaddi}) = (10 + 8)/40 = 9/20$ [1]
(iii) $P(\text{not football}) = 1 - 10/40 = 3/4$ [1]

- (iv) Neither cricket nor football: $40 - 26 = 14$, $P = 14/40 = 7/20$ [1]
37. (i) Differences $60 - 50 = 10$ and $70 - 60 = 10$ are equal, so it is an A.P.; $d = 10$ [1]
- (ii) $t_8 = 50 + 7 \times 10 = ₹120$ [1]
- (iii) $S_{12} = 12/2 [2 \times 50 + 11 \times 10] = 6 \times 210 = ₹1260$ [1]
- (iv) $200 = 50 + (n - 1) \times 10$, so $n - 1 = 15$ and $n = 16$: the 16th week [1]
38. (i) Circumference $= 2 \times 22/7 \times 14 = 88$ m [1]
- (ii) Area of park $= 22/7 \times 14 \times 14 = 616$ m² [1]
- (iii) Outer radius $= 14 + 3.5 = 17.5$ m; outer area $= 22/7 \times 17.5 \times 17.5 = 962.5$ m²; path area $= 962.5 - 616 = 346.5$ m² [1]
- (iv) Cost $= 346.5 \times 40 = ₹13,860$ [1]